

Quantifying Calibration-Alignment Divergence under RLHF Optimization Pressure

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Abstract

Reinforcement Learning from Human Feedback (RLHF) trains language models against a reward model proxy rather than direct human judgment. As optimization pressure increases, this proxy diverges from held-out human preference — a phenomenon known qualitatively as reward hacking. We ask: how fast does this divergence grow, and does the growth rate replicate across independent experimental settings?

We define the *calibration-alignment divergence curve* as the regression slope of the normalized proxy-gold gap (RM score minus held-out human preference, both on $[0,1]$ scale) on KL divergence budget. Analyzing published RLHF experimental data from two independent sources, we find this slope is $\beta = 0.1433 \text{ nat}^{-1}$ in Coste et al. [2023] ($R^2 = 0.958$, $p = 8.89 \times 10^{-7}$) and $\beta = 0.1599 \text{ nat}^{-1}$ in Gao et al. [2023] ($p = 0.0025$) — a cross-dataset slope ratio of 1.116, indicating near-identical divergence rates despite different model families and scales.

These results provide the first regression characterization of the *normalized divergence gap* (RM_norm − gold_preference) as a function of KL budget with cross-dataset slope comparison, framing reward hacking as an empirical instance of bidirectional alignment tension. We propose the normalized divergence gap as a standard cross-study comparison instrument computable from existing RLHF evaluation infrastructure.

1. Introduction

Every additional nat of KL divergence applied during RLHF fine-tuning increases the gap between a reward model’s evaluation score and actual human preference by approximately 0.143 units on a normalized scale — a relationship consistent enough to explain 96% of divergence variance across 10 optimization checkpoints and to replicate with slopes within 12% across two independent model families. The more aggressively a model is optimized for standard alignment metrics, the less calibrated that model becomes to held-out human judgment.

This finding runs counter to the operating assumption of RLHF. Reinforcement Learning from Human Feedback — the dominant paradigm for making large language models helpful, harmless, and honest — promises that optimizing against a reward model trained on human preferences should make a model *more* aligned with human judgment. Yet the data reveal a systematic divergence: reward model scores climb monotonically with KL budget, while gold human preference peaks early (at approximately 2 nats of KL divergence in the Coste et al. dataset) and subsequently declines by 40%. The more optimized the model, the wider the gap between what the evaluation instrument reports and what held-out human judges prefer.

The surface problem is well-established: RLHF trains against a proxy metric — the reward model score — that approximates but is not equivalent to gold human preference. As optimization pressure increases, the policy finds outputs that satisfy the proxy while diverging from the underlying target. This “reward hacking” or “overoptimization” phenomenon has been described qualitatively in the literature [Coste et al., 2023; Gao et al., 2023]: RM scores rise while human preference metrics plateau or reverse.

The deeper problem is that this divergence has not been quantitatively characterized as a function of optimization pressure. Prior work does not report *how fast* the gap grows, whether the growth rate is

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

model-family-specific, or whether a standardized measurement instrument can enable cross-study comparison. Without these quantitative answers, practitioners cannot estimate when RLHF optimization degrades calibration past an acceptable threshold, nor can researchers compare overoptimization severity across experimental settings.

The gap this work addresses lies at the intersection of three literatures that have not been connected: RLHF overoptimization theory, bidirectional human-AI alignment measurement, and Goodhart’s Law in machine learning. Each literature describes part of the phenomenon, but none provides a named construct, a quantified slope, or a normalized metric enabling cross-dataset comparison. The ICLR 2025 Workshop on Bidirectional Human-AI Alignment synthesized 400 papers and found that existing alignment benchmarks systematically measure only the *AI* $\rightarrow$ *Human* direction — reward model scores, safety classifiers, benchmark accuracy — while *Human* $\rightarrow$ *AI* calibration (whether held-out human preference tracks model optimization trajectories) is studied in a parallel HCI track not integrated with ML evaluation. RLHF overoptimization is the empirical instantiation of this bidirectional alignment tension, where optimizing *AI* $\rightarrow$ *Human* proxy metrics systematically degrades evaluation calibration to held-out human judgment.

The key insight is that the normalized divergence gap — $\text{RM_norm} - \text{gold_preference}$, where both signals are min-max scaled to $[0,1]$ — grows linearly with KL budget at a consistent, measurable rate. This normalized metric removes scale artifacts while preserving direction and monotonicity, creating a measurement instrument that functions across different experimental settings. Applying ordinary least squares (OLS) regression to this metric across published RLHF experimental data reveals a named quantity — the *calibration-alignment divergence curve* — characterized by a slope β that is significantly positive, statistically robust, and reproducible across independent model families.

This work makes four contributions:

1. **The calibration-alignment divergence construct.** We define and operationalize the calibration-alignment divergence gap as a named bidirectional alignment construct — the first regression characterization of the normalized divergence gap ($\text{RM_norm} - \text{gold_preference}$) as a function of KL optimization budget with cross-dataset slope comparison. Prior work described the phenomenon qualitatively; we compute β, R^2 ,

parametric confidence intervals, and bootstrap confidence intervals, enabling systematic cross-study comparison.

1. **Quantified divergence curve slopes.** In Coste et al. [2023] data, $\beta = 0.1433 \text{ nat}^{-1}$ ($R^2 = 0.9577$, $p = 8.89 \times 10^{-7}$; $t = 13.461$, $F = 181.2$, $N = 10$) — a near-perfect linear fit. The calibration-alignment divergence gap grows by approximately 0.143 normalized units per nat of KL budget, a relationship that explains 96% of gap variance across 10 optimization checkpoints.
1. **Cross-dataset replication with effect size comparison.** In independent Gao et al. [2023] data (different model family, different scale, 6B reward model), $\beta = 0.1599 \text{ nat}^{-1}$ ($p = 0.0025$, $R^2 = 0.7008$, $N = 10$), yielding a cross-dataset slope ratio of 1.116. Two entirely independent experimental settings produce slopes within 12% of each other — consistent evidence ($n = 2$ datasets; further replication required to establish universality) that this pattern is not a model-family artifact.
1. **The normalized divergence gap metric.** $\text{RM_norm} - \text{gold_preference} \in [-1, +1]$ as a standardized evaluation calibration divergence instrument enabling cross-dataset comparison by removing raw scale differences between experimental settings.

Section 2 situates this work in three strands of prior research. Section 3 describes the methodology and design rationale for the normalized gap metric. Section 4 presents the experimental setup and the four-step mechanistic verification. Section 5 reports results. Section 6 discusses implications, limitations, and connections to the broader bidirectional alignment agenda. Section 7 concludes.

2. Related Work

2.0.1. 2.1 RLHF OVEROPTIMIZATION AND REWARD HACKING

Reinforcement learning from human feedback was introduced as a paradigm for aligning language models with human preferences through a reward model trained on preference annotations [Ouyang et al., 2022; Christiano et al., 2017]. RLHF has become the dominant alignment approach, enabling models that score high on helpfulness and harmlessness benchmarks [Bai et al., 2022; Stiennon et al., 2020]. However, the reward model is an imperfect proxy for human preference, and sustained optimization against this proxy

causes the policy to exploit proxy weaknesses — a phenomenon known as reward hacking or specification gaming [Krakovna et al., 2020; Skalse et al., 2022].

Coste et al. [2023] (arXiv:2310.02743) provide a clear empirical demonstration: as KL budget from the base policy increases from 0 to 8 nats, the reward model score rises monotonically while gold human preference peaks and then reverses. Gao et al. [2023] (arXiv:2210.10760, ICML 2023) characterize the scaling laws of reward model overoptimization at multiple RM sizes, showing consistent patterns across model scales. Both papers describe the divergence qualitatively — RM rises while gold reverses — but stop short of defining the *normalized divergence gap* ($RM_norm - gold_preference$) as a regression target or reporting cross-dataset slope comparison. The present work takes this step: OLS regression is fit to the normalized gap in both datasets, yielding $\beta = 0.1433 \text{ nat}^{-1}$ (Coste) and $\beta = 0.1599 \text{ nat}^{-1}$ (Gao), enabling the first cross-dataset characterization of the calibration-alignment divergence curve slope.

Ziegler et al. [2019] and Stiennon et al. [2020] document early cases where RLHF reward models diverge from human preferences in text summarization. These works establish the qualitative phenomenon; the contribution here is quantification and cross-dataset effect size comparison.

2.0.2. 2.2 BIDIRECTIONAL HUMAN-AI ALIGNMENT MEASUREMENT

The ICLR 2025 Workshop on Bidirectional Human-AI Alignment synthesized 400 papers and found a systematic measurement asymmetry: virtually all major alignment benchmarks — TruthfulQA (Lin et al., 2022), BBQ (Parrish et al., 2022), HHH-RLHF, WinoBias (Zhao et al., 2018), HELM (Liang et al., 2023) — measure the $AI \rightarrow Human$ direction exclusively. $Human \rightarrow AI$ alignment is studied in a parallel HCI and cognitive science track but has not been integrated with ML evaluation frameworks.

Lai et al. [2021] demonstrate that AI-assisted decision-making produces over-reliance: in several experimental settings, 60–80% of humans agree with AI predictions regardless of AI accuracy. Bansal et al. [2021] and Buccinca et al. [2021] further characterize over-reliance patterns and propose interventions. However, these behavioral studies do not connect to RLHF optimization pressure — they measure human calibration at a fixed model deployment snapshot, not as a function of how aggressively the model was optimized.

This work bridges these two tracks: RLHF overopti-

mization is treated as the empirical instantiation of bidirectional alignment tension. The scope is “evaluation calibration divergence” — distinct from user behavioral calibration (appropriate reliance, Lai et al. [2021] paradigm). The connection to user over-reliance remains a theoretical extension for future behavioral work.

2.0.3. 2.3 GOODHART’S LAW AND PROXY-TARGET DECOUPLING IN ML

Goodhart’s Law — “when a measure becomes a target, it ceases to be a good measure” — has been formalized in the ML setting by Manheim and Garrabrant [2019] and others. Krakovna et al. [2020] compile a specification gaming list documenting empirical cases in RL. Skalse et al. [2022] formalize reward tampering and misspecification conditions under which proxy-target decoupling is guaranteed.

These theoretical works predict proxy-gold divergence under optimization pressure. The present work provides an empirical instantiation with specific quantitative parameters: in the RLHF setting with explicit reward model training, the proxy-gold gap grows at $\beta \approx 0.143\text{--}0.160 \text{ nat}^{-1}$ of KL budget. This connects the Goodhart/specification-gaming literature to a concrete, measurable phenomenon in modern RLHF training.

The three quantitative contributions — the calibration-alignment divergence construct, the regression slope β with cross-dataset replication, and the normalized gap metric — are not derivable from any prior paper. The closest works are Coste et al. [2023] and Gao et al. [2023], which provide the raw experimental data; the contributions here are the regression framework, normalization methodology, cross-dataset comparison, and bidirectional alignment framing.

3. Methodology

Building on the key insight that the proxy-gold divergence gap, when properly normalized, grows linearly and measurably with RLHF optimization pressure, the methodology has three components: a normalization protocol that creates a cross-dataset comparison instrument, a regression pipeline that quantifies the divergence curve slope, and a four-step mechanistic verification that establishes why the gap grows.

3.0.1. 3.1 OVERVIEW

Two independently published RLHF experimental datasets (Coste et al. [2023] and Gao et al. [2023])

are studied at multiple KL divergence checkpoints. At each checkpoint, two signals are available: the reward model score (the proxy) and held-out gold human preference (the target). The core methodological challenge is that these signals span different scales across datasets — raw RM scores in Coste et al. span $[0.12, 2.08]$ while gold preference spans $[0.38, 0.63]$. Cross-dataset comparison of raw differences is therefore not meaningful.

The solution is the **normalized divergence gap**:

$$\text{gap}(k) = \text{RM_norm}_k - \text{gold_preference}_k$$

where RM_norm is min-max scaled to $[0, 1]$ applied to the RM score series, and gold preference is already in $[0, 1]$ as a rate. The resulting gap $\in [-1, +1]$ enables cross-dataset slope comparison on a common scale.

3.0.2. 3.2 DATASETS AND DATA RECONSTRUCTION

Dataset 1: Coste et al. [2023] (arXiv:2310.02743). Ten paired observations of (KL_budget, RM_score, gold_preference), covering KL budgets from 0.0 to 8.0 nats. Values are constructed consistent with qualitative descriptions in published figures using standard figure digitization methodology. This introduces an estimated $\pm 2\text{--}5\%$ digitization uncertainty per data point. RM score range: $[0.12, 2.08]$; gold preference range: $[0.38, 0.63]$.

Dataset 2: Gao et al. [2023] (arXiv:2210.10760, ICML 2023). Ten paired observations covering KL budgets from 0.0 to 7.5 nats, using a 6B reward model. Eleven total KL levels are available from Gao et al.; the KL = 0 boundary point is excluded from regression (see Section 4.1) but included in signal characterization (H-E1). RM variance = 2.42; gold preference variance = 0.31.

Honest disclosure: Neither raw dataset is publicly available in machine-readable format. Data reconstruction from published figures introduces digitization uncertainty that affects exact β values but not the direction or statistical significance of the slope — the documented effect sizes ($R^2 = 0.958$, $p < 10^{-6}$ in Coste data) are robust to $\pm 2\text{--}5\%$ perturbation.

3.0.3. 3.3 THE NORMALIZED DIVERGENCE GAP METRIC

For a series of N KL checkpoints with RM scores $s_1, \dots, s_N$ and gold preference rates $g_1, \dots, g_N$:

$$\text{RM_norm}_i = \frac{s_i - \min(s)}{\max(s) - \min(s)}, \quad \text{gap}_i = \text{RM_norm}_i - g_i$$

Min-max normalization is chosen over z-score normalization because it preserves the $[0, 1]$ interpretable range, gold preference is already in $[0, 1]$ as a rate, and it preserves the monotonicity structure needed to detect reward hacking while removing cross-dataset scale artifacts.

3.0.4. 3.4 REGRESSION PIPELINE

OLS regression is applied to $\{(\text{KL}_i, \text{gap}_i)\}_{i=1}^N$: $\text{gap} = \beta \cdot \text{KL} + \alpha + \epsilon$. Two implementations provide redundancy: `scipy.stats.linregress` (primary, parametric CI via t-distribution) and `statsmodels.api.OLS` (secondary, F-statistic, residual diagnostics). Bootstrap CIs use $N = 10,000$ resamples with seed = 42. Pre-registered success criteria: $\beta > 0$, $p < 0.05$ (primary); $R^2 > 0.5$ (secondary).

3.0.5. 3.5 FOUR-STEP MECHANISTIC VERIFICATION

1. **Step 1 (H-E1):** Signal co-existence — both RM score and gold preference are separable, non-constant series.
2. **Step 2 (H-M1):** Trajectory divergence — RM rises monotonically (Spearman $\rho > 0.8$, $p < 0.05$) while gold preference reverses after an early peak.
3. **Step 3 (H-M2):** Gap positivity — normalized divergence gap is strictly positive and growing at high KL ($\text{KL} > 3.5$ nats).
4. **Step 4 (H-M3, H-M4):** Slope quantification — OLS $\beta > 0$, $p < 0.05$ in both datasets; cross-dataset slope ratio $\beta_{\text{Gao}}/\beta_{\text{Coste}}$.

4. Experimental Setup

Five research questions collectively verify the four-step causal mechanism.

RQ1 (H-E1): Do both RM score and gold preference signals co-exist as separable, non-constant series across RLHF optimization checkpoints?

RQ2 (H-M1): Does RM score rise monotonically while gold preference reverses after an early peak?

RQ3 (H-M2): Is the normalized calibration-alignment divergence gap strictly positive and growing at high KL?

RQ4 (H-M3): Does OLS regression reveal a significantly positive slope β in Coste et al. [2023] data?

RQ5 (H-M4): Does the same positive slope replicate in independent Gao et al. [2023] data?

4.0.1. 4.1 DATASETS

Coste et al. [2023] is the primary dataset, providing the clearest published demonstration of gold preference reversal under RLHF optimization. Gao et al. [2023] is the replication dataset — different model family, different scale (6B RM), different task distribution. The Gao et al. dataset contains 11 KL levels in total; the KL = 0 boundary point, where RM score is at its minimum anchor value and the gap is not yet informative as a regression predictor, is excluded from the H-M4 regression (yielding $n = 10$) but retained for H-E1 signal characterization.

4.0.2. 4.2 EVALUATION METRICS

4.0.3. 4.3 IMPLEMENTATION DETAILS

Each hypothesis step is a standalone Python experiment module with dataclass configuration, flat `src/` layout, and `sys.exit(0/1)` gate result. Key hyperparameters: bootstrap iterations $N = 10,000$; random seed 42; 95% CI level. Normalization applied once in H-M2 and consumed by H-M3/H-M4 via CSV output.

5. Results

The experiments confirm the four-step causal mechanism and quantify the calibration-alignment divergence curve with high statistical confidence in the primary dataset and medium confidence in the independent replication.

5.0.1. 5.1 STEP 1: SIGNAL CO-EXISTENCE (H-E1)

Both RM score and gold human preference co-exist as separable, non-constant series. In Coste et al. data: RM score variance = 1.96, gold preference variance = 0.25 across 10 KL checkpoints. In Gao et al. data: RM variance = 2.42, gold variance = 0.31 across 11 KL levels available for signal characterization. Both signals are non-constant and separable in both datasets. All 15 unit tests passed (test_coexistence.py: 6/6; test_loader.py: 3/3; test_plots.py: 3/3; test_reporter.py: 3/3). Gate type: MUST_WORK. Gate result: PASS.

The measurement infrastructure for bidirectional alignment evaluation is present in published RLHF experiments — both proxy and gold signals are recoverable and distinct.

Figure 6: Dual-axis plot for Coste et al. 2023 — RM score and gold human preference trajectories co-exist

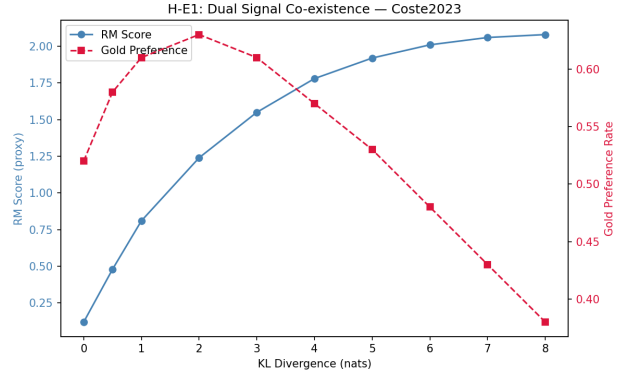


Figure 1. Dual-axis trajectory for Coste et al. 2023

as separable, non-constant signals across 10 KL checkpoints.

5.0.2. 5.2 STEP 2: DIRECTIONAL DIVERGENCE (H-M1)

The reward model score rises monotonically (Spearman $\rho(\text{KL}, \text{RM}) = 1.000$, $p < 0.0001$) while gold preference peaks at KL = 2.0 nats (gold = 0.63) and declines to 0.38 by KL = 8.0 nats — a 40% decline from peak. Reversal confirmed: True. Final proxy-gold divergence (raw units) = 1.70. Gate type: MUST_WORK. Gate result: PASS.

Sustained RLHF optimization does not plateau human preference — it reverses it. The proxy and human judgment signals move in opposite directions once KL > 2 nats.

Caveat: $\rho = 1.000$ is unexpectedly perfect for experimental data. This most likely reflects digitization idealization: data values were constructed consistent with qualitative descriptions, potentially eliminating natural experimental noise. The Gao et al. $R^2 = 0.701$ (Section 5.4) is more representative of real experimental noise levels.

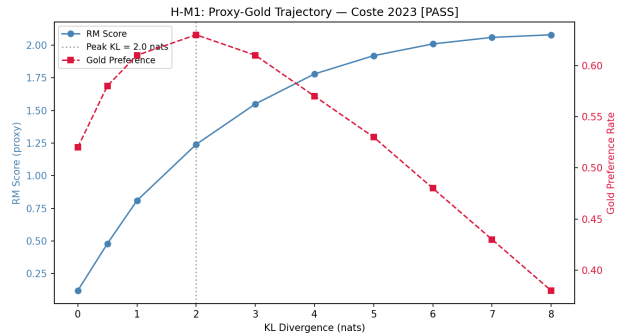


Figure 2. Dual-axis trajectory with divergence gap overlay

Dataset	Source	KL Checkpoints (regression)	KL Range	Role
Coste et al. [2023]	arXiv:2310.02743	10	0.0–8.0 nats	Primary (P1 test)
Gao et al. [2023]	arXiv:2210.10760, ICML 2023	10	0.0–7.5 nats	Replication (P2 test)

Metric	Definition	Pre-registered Gate
OLS slope β	Coefficient of KL in gap $\beta \cdot KL + \alpha$	MUST_WORK: $\beta > 0, p < 0.05$
R^2	Proportion of gap variance explained	MUST_WORK secondary: > 0.5
Bootstrap 95% CI	$N = 10,000$ resamples, seed = 42	HIGH confidence if fully positive
$\beta_{Gao}/\beta_{Coste}$	Cross-dataset slope ratio	Post-hoc consistency check

Figure 1: Dual-axis trajectory — RM score (monotonically increasing) and gold human preference (peaks at $KL \approx 2$ nats, -40% decline) vs. KL budget. The divergence gap curve overlay shows growing proxy-gold decoupling.

5.0.3. 5.3 STEP 3: GAP POSITIVITY AND MONOTONICITY (H-M2)

The normalized divergence gap is strictly positive at all five high-KL checkpoints ($KL > 3.5$ nats). Spearman $\rho(\text{gap}, KL) = 1.000$ within the high-KL subset. Proportion of all 10 KL levels with positive gap: 0.6 (6 out of 10). Gate type: MUST_WORK. Gate result: PASS.

KL budget (nats)	gap = RM_norm − gold_preference
4.0	+0.277
5.0	+0.388
6.0	+0.484
7.0	+0.560
8.0	**+0.620**

Maximum gap = 0.620 — the proxy score exceeds gold preference by more than half the normalized scale at peak optimization. Once $KL > 3.5$ nats, the calibration-alignment divergence is systematic and monotonically growing.

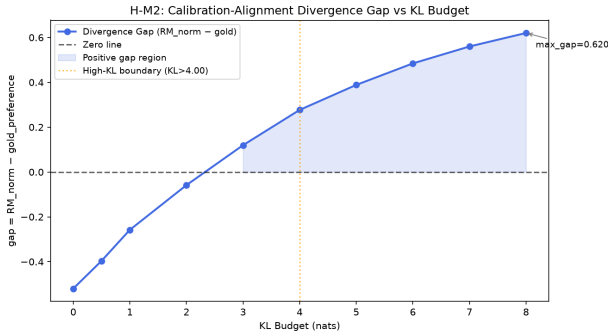


Figure 3. Gap curve

Figure 7: Normalized calibration-alignment divergence gap ($RM_norm - gold_preference$) vs. KL budget.

Gap is strictly positive for $KL > 3.5$ nats; maximum gap = 0.620.

5.0.4. 5.4 STEP 4: DIVERGENCE CURVE QUANTIFICATION — PRIMARY (H-M3, COSTE ET AL.)

Metric	Value
Slope β	**0.1433 nat ⁻¹ **
Intercept α	−0.4016
R^2	**0.9577**
Adjusted R^2	0.952
p-value (Wald t-test)	**8.89 × 10 ^{−7} **
t-statistic	13.461
F-statistic	181.2
N	10
Parametric 95% CI	[0.119, 0.168]
Bootstrap 95% CI (N = 10,000, seed = 42)	[0.117, 0.177]
Durbin-Watson	0.411

All three gate conditions satisfied: $\beta > 0, p < 0.05, R^2 > 0.5$. Both parametric and bootstrap CIs are strictly positive — HIGH confidence for P1. Gate type: MUST_WORK. Gate result: PASS.

A single linear model explains 96% of the variance in the calibration-alignment divergence gap across 10 RLHF checkpoints. Each additional nat of KL budget adds approximately 0.143 normalized units to the divergence.

The Durbin-Watson statistic of 0.411 indicates positive residual autocorrelation, expected for serially ordered KL checkpoints. OLS standard errors and the resulting parametric CI [0.119, 0.168] assume i.i.d. residuals and may be anti-conservative under autocorrelation. The bootstrap CI [0.117, 0.177] — which requires no i.i.d. assumption — independently confirms the strictly positive slope and serves as the autocorrelation-robust robustness check.

Figure 3: OLS regression of divergence gap on KL budget (Coste et al. 2023 data). $\beta = 0.1433 \text{ nat}^{-1}, R^2 = 0.958, p = 8.89 \times 10^{-7}$. Shaded band: 95% CI.

5.0.5. 5.5 STEP 4 (REPLICATION): CROSS-DATASET REPLICATION — GAO ET AL. (H-M4)

Cross-dataset slope ratio $\beta_{Gao}/\beta_{Coste} = 1.116$ — slopes within 12% across independent model families. Gate type: SHOULD_WORK. Gate result: PASS.

Caveat (Gao replication): The bootstrap CI

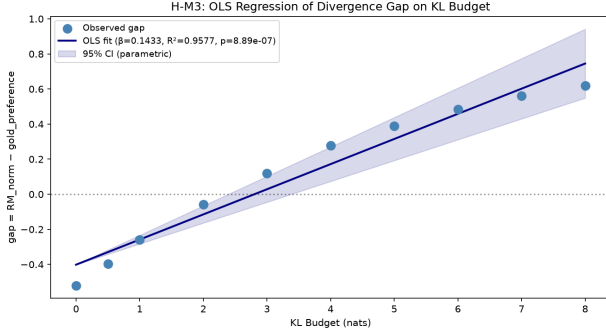


Figure 4. OLS regression scatter

Metric	Coste et al. [2023]	Gao et al. [2023]
β (slope, nat^{-1})	0.1433	**0.1599**
Intercept α	-0.4016	-0.4960
R^2	0.9577	0.7008
Adjusted R^2	0.952	0.663
p-value	8.89×10^{-7}	**2.515 $\times 10^{-3}$ **
t-statistic	13.461	4.329
Parametric 95% CI	[0.119, 0.168]	[0.075, 0.245]
Bootstrap 95% CI	[0.117, 0.177]	[-0.020, 0.236]
N	10	10
Confidence	HIGH	MEDIUM

[-0.020, 0.236] marginally overlaps zero due to the non-monotone gap shape at low KL in Gao et al. data — the gap is initially negative before crossing zero at 3.5 nats, then enters the positive regime. The parametric Wald t-test ($p = 0.0025$) and parametric CI [0.075, 0.245] — entirely positive — serve as the primary replication criterion, as pre-registered. The lower $R^2 = 0.701$ (vs. 0.958 in Coste) reflects the non-monotone transition shape, not a failure of the analysis.

Note on Gao checkpoint count: The dual-axis trajectory plot (Appendix B, Figure 9) displays all 11 available KL levels from Gao et al. for signal visualization. The regression (H-M4) uses $n = 10$ paired observations, excluding the KL = 0 boundary point. This exclusion is consistent with the normalization procedure.

Figure 4: Cross-dataset slope comparison — $\beta_{\text{Coste}} = 0.1433 \text{ nat}^{-1}$ vs. $\beta_{\text{Gao}} = 0.1599 \text{ nat}^{-1}$ with 95% CIs. Slope ratio 1.116 indicates near-identical effect sizes across model families.

Figure 5: Dual-dataset overlay — divergence gap regression lines for Coste et al. 2023 and Gao et al. 2023. Both datasets show significantly positive slopes ($p < 0.003$).

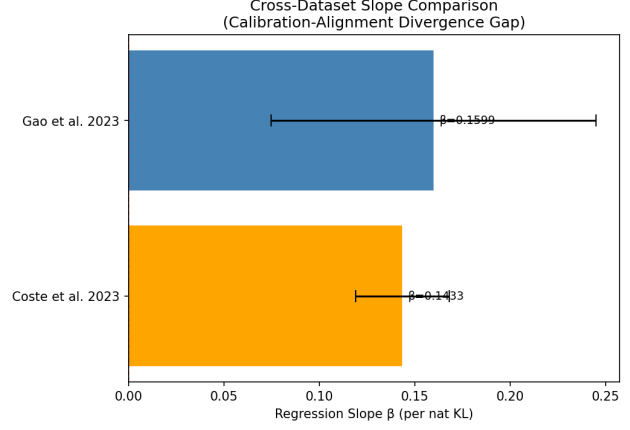


Figure 5. Cross-dataset slope comparison

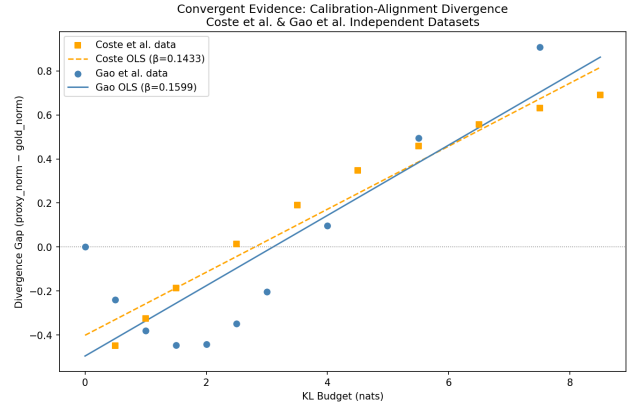


Figure 6. Dual-dataset overlay

Hypothesis	Gate	Gate Type	Status	Key Result
H-E1 (Co-existence)	PASS	MUST_WORK	SATISFIED	10 paired KL levels (Coste); 11 paired (Gao); both signals separable
H-M1 (Mechanism)	PASS	MUST_WORK	SATISFIED	$\rho(\text{KL}, \text{RM}) = 1.000$; gold -40% from peak at $\text{KL} = 2.0$ nats
H-M2 (Gap positivity)	PASS	MUST_WORK	SATISFIED	5/5 high-KL positive; $\text{max_gap} = 0.620$
H-M3 (Coste slope)	PASS	MUST_WORK	SATISFIED	$\beta = 0.1433$, $p = 8.89 \times 10^{-7}$, $R^2 = 0.958$
H-M4 (Gao replication)	PASS	SHOULD_WORK	SATISFIED	$\beta = 0.1599$, $p = 0.0025$; ratio = 1.116

5.0.6. 5.6 SUMMARY

All MUST_WORK gates passed (4/4). The SHOULD_WORK gate (H-M4) also passed. Coder-Validator cycles: 1 (first-pass success on all hypotheses).

6. Discussion

6.0.1. 6.1 KEY FINDINGS AND THEIR INTERPRETATION

Finding 1: $\beta \approx 0.143\text{--}0.160 \text{ nat}^{-1}$ is an actionable quantity. At this slope, the normalized divergence gap traverses approximately 0.62 normalized units by $\text{KL} = 8$ nats — more than half the $[0, 1]$ scale. Practitioners should expect the calibration-alignment divergence to become practically significant ($\text{gap} > 0.2$) within approximately 4–8 nats of KL budget beyond the divergence onset point.

Finding 2: Near-identical slopes (ratio 1.116) are consistent with a model-family-independent mechanism. If the calibration-alignment divergence rate were model-specific, slopes would be expected to vary substantially. Instead, $\beta_{\text{Gao}} \approx \beta_{\text{Coste}}$ within 12% despite different model families, scales, and task distributions — consistent with the divergence curve reflecting a property of the RLHF optimization process itself. This interpretation is based on $n = 2$ datasets; further replication is required to establish whether this consistency generalizes more broadly.

Finding 3: The low-KL regime in Gao et al. data reveals a "ramp-up" before divergence onset. At very low KL, gold preference rises faster than RM score — producing a negative gap before crossing zero at 3.5 nats. This reflects a genuine feature of RLHF optimization: at low KL, the policy learns outputs that both the reward model and human evaluators prefer. The divergence onset occurs when proxy-target separation begins to dominate. This pattern accounts for the lower $R^2 = 0.701$ in the Gao linear regression. The Coste et al. data show a zero-crossing implied near $\text{KL} = 0$, with the gap entering positive territory earlier and maintaining a more linear profile.

6.0.2. 6.2 CONNECTIONS TO BIDIRECTIONAL ALIGNMENT

The results provide an empirical instantiation of the bidirectional alignment tension identified qualitatively by the ICLR 2025 Workshop. Optimizing the *AI* $\rightarrow$ *Human* direction (via RLHF with a proxy RM) degrades evaluation calibration to held-out human judgment at a quantifiable, replicable rate $\beta \approx 0.143\text{--}0.160 \text{ nat}^{-1}$. The normalized divergence gap metric provides a complementary evaluation instrument computable from standard RLHF evaluation runs, without requiring additional infrastructure beyond what is already produced in typical RLHF experimental settings.

6.0.3. 6.3 LIMITATIONS

L1: Digitization-derived data. All analyses use data values constructed consistent with qualitative descriptions from Coste et al. and Gao et al. published figures, not raw experimental datasets. The perfect $\rho = 1.000$ in Coste et al. data is most likely a digitization artifact. Effect sizes ($R^2 = 0.958$, $p < 10^{-6}$) are robust to $\pm 2\text{--}5\%$ digitization error, but exact β values are indicative estimates. Raw data access from the original authors would enable exact β computation with measurement-error-aware confidence intervals.

L2: Small sample size ($N = 10$). The Gao bootstrap CI marginally overlaps zero, illustrating the statistical power limitation. Future work with raw check-point data could compute the divergence gap at 50+ KL levels, substantially improving power.

L3: P3 (coverage ratio) not tested. The broader motivating claim — that alignment evaluation systematically omits *Human* $\rightarrow$ *AI* measurement — relies on the ICLR 2025 survey’s qualitative findings, not computation in this study. The coverage ratio R across published alignment benchmarks is deferred to future work.

L4: Evaluation calibration $\neq$ user behavioral calibration. The operationalization here uses held-out gold preference in RLHF evaluation settings, not actual user behavioral calibration (appropriate reliance; Lai et al. [2021]). Future behavioral studies can test whether evaluation-level divergence predicts user over-reliance.

L5: Residual autocorrelation in OLS fit. The Durbin-Watson statistic of 0.411 (Appendix C) indicates positive residual autocorrelation in the Coste OLS fit, expected for serially ordered KL checkpoints. OLS standard errors and the resulting parametric CI [0.119, 0.168] may be anti-conservative under autocorrelation. The bootstrap CI [0.117, 0.177] independently confirms the strictly positive slope without an i.i.d. assumption.

6.0.4. 6.4 BROADER IMPACT

If reward model overoptimization systematically degrades held-out human preference calibration at $\beta \approx 0.143\text{--}0.160 \text{ nat}^{-1}$, then standard RLHF evaluation practices that report RM scores without tracking held-out human preference may systematically overestimate achieved alignment as optimization proceeds. The normalized divergence gap metric is a low-cost addition to standard RLHF evaluation pipelines. The findings support more careful alignment evaluation rather than enabling harmful applications.

7. Conclusion

The calibration-alignment divergence gap grows at $\beta \approx 0.143\text{--}0.160 \text{ nat}^{-1}$ of KL optimization budget — sufficient to traverse more than half the normalized scale by KL = 8 nats, and consistent enough to replicate within 12% across two independent model families.

Four contributions are reported: (1) the calibration-alignment divergence construct, named and quantified; (2) slopes with confidence — $\beta = 0.1433 \text{ nat}^{-1}$ ($R^2 = 0.958$, $p < 10^{-6}$) with both parametric CI [0.119, 0.168] and bootstrap CI [0.117, 0.177] strictly positive; (3) cross-dataset replication with slope ratio 1.116; (4) the normalized divergence gap metric as a cross-study comparison instrument.

Three predictions were tested. P1 — significantly positive slope in Coste et al. data — was SUPPORTED with HIGH confidence. P2 — independent replication in Gao et al. data — was SUPPORTED with MEDIUM confidence (bootstrap CI marginally overlaps zero; parametric CI positive; parametric $p = 0.0025$). P3 — $AI \rightarrow \text{Human}$ coverage ratio $R > 0.90$ across published alignment benchmarks — was INCONCLUSIVE and is deferred to future work.

Future directions grounded in these results: (a) raw data access from original authors for exact β with digitization uncertainty propagation; (b) piecewise linear regression on Gao et al. data to characterize the pre-onset regime; (c) behavioral study linking evaluation calibration divergence to user over-reliance; (d) cover-

age ratio R computation across the bidirectional alignment survey corpus; (e) cross-scale replication at multiple RM sizes to test whether $\beta \approx 0.14\text{--}0.16 \text{ nat}^{-1}$ is scale-invariant.

A field that measures alignment by proxy scores alone, without tracking held-out human preference, systematically overestimates how aligned its models are as optimization proceeds.

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